

LYAKHSH, Tajikistan

WMO: 387440

Lat: 38.317N

Lon: 71.267E

Elev: 1685

StdP: 82.65

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-16.5	-23.1	0.6	-18.1	-20.7	0.7	-15.9	7.4	-3.9	6.1	-3.2	0.3	N/A	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	29.9	15.0	28.4	14.2	27.0	13.7	16.0	26.7	15.2	25.8	14.5	24.9	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	11.1	19.3	11.3	10.2	18.3	10.3	9.6	17.5	51.3	26.7	48.8	25.8	46.5	24.9	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	5.9	4.6	-21.4	32.5	3.8	1.5	-24.1	33.6	-26.4	34.5	-28.5	35.3	-31.3	36.4		
(5)			-21.9	17.9	3.6	0.9	-24.5	18.6	-26.6	19.1	-28.6	19.6	-31.2	20.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-7.7	-5.7	1.0	8.8	12.9	16.1	19.2	19.1	15.4	9.0	2.1	-4.4
	DBStd	10.05	4.94	4.82	4.87	3.68	2.54	2.47	2.58	2.53	2.84	3.19	3.81	3.85
	HDD10.0	2079	548	441	278	64	5	1	0	0	2	58	237	446
	HDD18.3	4157	806	674	536	286	169	74	20	19	92	290	486	704
	CDD10.0	1064	0	0	1	28	94	183	286	282	164	26	0	0
	CDD18.3	101	0	0	0	0	0	6	48	43	4	0	0	0
	CDH23.3	1688	0	0	0	2	19	158	667	672	167	3	0	0
	CDH26.7	381	0	0	0	0	1	15	179	170	17	0	0	0
Wind	WSAvg	1.1	0.5	0.6	0.8	1.4	1.4	1.3	1.4	1.6	1.5	1.2	0.9	1.0
Precipitation	PrecAvg	527	51	64	76	68	59	33	27	16	10	31	43	50
	PrecMax	659	103	108	119	110	120	60	72	28	19	80	111	96
	PrecMin	389	23	25	27	28	30	16	10	7	1	11	14	17
	PrecStd	74	20	17	23	20	24	12	12	6	5	16	24	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	8.1	15.5	22.9	25.6	28.5	32.1	32.1	28.5	23.3	16.0	6.2
		MCWB	-0.6	2.6	6.8	10.6	13.0	14.7	15.8	15.2	13.3	10.1	6.5	1.2
	2%	DB	1.1	3.6	12.6	20.0	23.3	26.5	30.3	30.0	26.8	20.5	12.9	3.9
		MCWB	-1.3	0.9	5.8	9.7	12.0	13.8	15.3	14.7	12.4	9.1	4.9	0.5
	5%	DB	0.0	1.8	9.9	18.0	21.6	25.2	28.6	28.5	25.3	18.3	10.5	2.4
		MCWB	-2.0	-0.3	4.9	8.6	11.3	13.3	14.7	13.9	11.8	8.2	3.8	-0.6
	10%	DB	-1.1	0.6	7.5	15.8	19.9	23.6	27.1	27.1	23.7	16.3	8.2	1.0
		MCWB	-2.8	-1.1	3.7	8.1	10.6	12.9	14.2	13.7	11.1	7.2	3.0	-1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	3.2	7.7	11.6	13.7	15.7	17.5	17.2	14.5	10.8	7.0	2.5
		MCDB	1.5	7.1	13.5	20.8	22.8	25.4	28.0	27.2	26.6	21.2	14.8	4.7
	2%	WB	-0.7	1.1	6.2	10.2	12.6	14.7	16.3	15.9	13.2	9.7	5.6	1.0
		MCDB	0.6	3.3	11.5	18.0	21.4	24.2	26.9	26.4	24.7	18.8	11.2	3.2
	5%	WB	-1.8	0.0	5.1	9.3	11.9	14.0	15.5	15.0	12.2	8.8	4.6	-0.2
		MCDB	-0.4	1.5	9.5	16.7	19.7	23.2	26.2	26.2	23.9	16.8	9.2	1.9
	10%	WB	-2.8	-1.0	3.9	8.4	11.1	13.3	14.8	14.2	11.4	7.8	3.5	-1.2
		MCDB	-1.1	0.5	7.4	14.9	18.4	22.1	25.1	25.1	22.3	14.7	7.5	0.8
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.5	8.2	10.2	11.8	13.5	14.6	15.4	15.3	12.8	9.4	7.7
		MCDBR	7.6	7.5	11.4	14.6	15.6	16.1	16.9	17.0	17.5	16.1	13.5	8.8
		MCWBR	5.8	5.5	6.0	6.3	6.1	5.8	5.4	5.8	6.4	6.8	7.1	6.2
	5% WB	MCDBR	6.1	6.8	10.2	12.9	13.7	14.6	15.3	15.2	16.5	14.4	11.6	7.4
		MCWBR	4.9	5.0	5.6	5.8	5.7	5.7	5.4	5.5	6.4	6.3	6.2	5.5
Clear-Sky Solar Irradiance	taub	0.328	0.373	0.449	0.476	0.483	0.489	0.481	0.433	0.384	0.395	0.367	0.318	
	taud	2.036	1.915	1.800	1.834	1.874	1.879	1.923	2.067	2.170	2.048	2.060	2.124	
	Ebn at Noon	817	824	796	805	808	801	804	835	857	794	763	802	
	Edh at Noon	122	159	198	203	199	199	188	159	135	137	119	104	
All-Sky Solar Radiation	RadAvg	2.52	3.34	4.72	5.73	6.19	6.98	7.26	7.07	5.92	3.78	2.44	2.17	
	RadStd	0.18	0.31	0.34	0.50	0.47	0.45	0.38	0.35	0.30	0.46	0.36	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page